

perhaps even cheaper in the long run because one can optimise home systems more easily with these.

Nine open source solutions available in the market, with their main features, are given below.

Home Assistant

This open source home automation platform was developed with software written in Python. It works asynchronously and has very low requirements in terms of specifications. It can control a home's lights and window blinds using a programmable logic controller connected to a home assistant using MQTT protocol. It can receive information from several sensors using a radio frequency (RF) gateway. It can also be connected to IP cameras to retrieve motion sensing events and send notifications through various services.

openHab

This home automation platform aims for integration and interaction between different home technologies and devices.

Main features are:

- Vendor- and technology-agnostic
- Developed in Java
- Runs on any device capable of running Java Virtual Machine (JVM)
- Integrates different home automation technologies into one
- Comes with different Web-based user interfaces (UIs)
- Provides APIs for being integrated with other systems

MyController

This home automation server allows monitoring and control of the home from anywhere.

Main features are:

- Lightweight server
- Supports Linux, Windows and Raspberry Pi
- Supports multiple gateways with many protocols (serial, Ethernet and MQTT)
- Time-based operations
- Gives notifications, alarms and graphical reports
- Comes with interactive dashboards

Domoticz

This is a lightweight home automation system that allows monitoring and configuration of lights, switches, sensors and other devices. It also allows you to send notifications and alerts to any mobile device.

Main features are:

- Compatible with all browsers
- Works on several operating systems with a scalable HTML 5 Web front-end that adapts to desktop and mobile devices
- Runs on Raspberry Pi, Cubie board, UNIX, Apple and Windows

Fhem

This is a Perl server for home automation, released under GPL licence. It automates tasks like switching on lamps, window shutters, heating and so on. It also logs stats like temperature, humidity and power consumption.

Main features are:

- Runs as a server
- Controls devices via Web or smartphone front-end, and via Telnet or TCP/IP directly
- Supports several protocols
- Auto creates a device's logs when receiving data from a new device
- Logs events to files or database
- Sends notifications about external programs or scripts when it receives data about certain events
- Timed commands
- Several interfaces: Simple text, JSON, XML, etc

Major DoMo

This is an open source home automation platform designed to be used in a multi-protocol and multi-services environment.

Main features are:

- Can be installed on almost any computer running Windows or Linux
- Easy installation
- Supports many protocols (MQTT, Zwave, I-wave, I-wore, ModBus, SNMP, Ethernet)
- Offers Web-based interfaces for any device
- GPS tracking
- Scripting with PHP and/or Blockly

- Sets up automation rules and scripts

Pytomatic

This is a lightweight open source automation system written in Python. The software can be used for home automation and lighting control, though it is not limited to these uses alone. It can work on any platform that supports Python (Windows, Mac OS X, Linux, etc).

Main features are:

- Executes actions based on voice inputs, time of day, file data, serial port data and socket data, as well as serial and speech protocols
- REST API
- Mobile Web and Android clients with continuous device updates (Web sockets)
- Voice commands from Android (Home Control app)
- Local Telnet and Web access
- Has a unique language to describe devices and actions
- Works with smart objects like doors, lights, motion sensors and photo cells
- Optional Mini loop programming for complicated controls
- Easy to add new hardware drivers
- Good communication, complete with examples
- Maps one command to another with source and time
- Optional event-driven programming for complex actions when the device state changes

Freedomotic

This open source, flexible, secure Internet of Things (IoT) development framework is useful to build and manage modern smart spaces. It is integrated with popular building automation technologies like BTicino Open Web Net, KNX Modbus RTU and X10, as well as with custom automation projects using Arduino devices, DIY boards, third-party Android front-ends, text-to-speech engines, motion-detection sensors using IP cameras, social network integration like Twitter and much more.

Freedomotic complies with well-known standards and building auto-